

Clinical Characteristics of Data Cohorts

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This report provides a comparative summary of patient demographic, treatment, survival, and sample attrition characteristics across the four cohorts analysed in this study:

- **TCGA-SKCM:** Baseline reference cohort with recorded treatment history.
- **Liu 2019:** Advanced melanoma trial cohort treated with anti-PD-1 monotherapy.
- **Hugo 2016:** Anti-PD-1 clinical trial cohort.
- **Riaz 2017:** Anti-PD-1 clinical trial cohort treated with nivolumab.

Demographic and treatment characteristics were calculated dynamically from the processed clinical datasets. Survival characteristics were calculated from cleaned clinical records evaluated in the Kaplan-Meier analysis. Sample attrition details track cohort retention across data preprocessing stages.

Table 1: Baseline Patient and Disease Characteristics

Characteristic	Liu 2019	Hugo 2016	Riaz 2017	TCGA-SKCM
N	122	27	107	443
Demographics				
Age, median (IQR)	N/A	61.0 (54.0–68.5)	56.0 (48.8–63.0)	57.0 (47.0–70.0)
Female sex, n (%)	51 (41.8%)	8 (29.6%)	47 (51.1%)	169 (38.1%)
Treatment				
ICI agent — Pembrolizumab	71 (58.2%)	0 (0.0%)	—	3 (0.7%)
ICI agent — Nivolumab	51 (41.8%)	0 (0.0%)	0 (0.0%)	1 (0.2%)
Prior anti-CTLA-4	0 (0.0%)	—	55 (100.0%)	—
Treatment History (TCGA only)	—	—	—	
— Radiation Therapy	—	—	—	112 (25.3%)
— Immunotherapy	—	—	—	70 (15.8%)
— Chemotherapy	—	—	—	67 (15.1%)
— Vaccine	—	—	—	22 (5.0%)
— Targeted Therapy	—	—	—	12 (2.7%)
— Other Therapy	—	—	—	7 (1.6%)
— No recorded treatment	—	—	—	241 (54.4%)
Survival Outcomes				
Median OS, months (95% CI)	22.6	32.2	21.2	79.0
OS events, n (%)	62 (50.8%)	12 (46.2%)	63 (62.4%)	212 (49.6%)
Median follow-up, months	17.5	14.4	17.8	41.6

Sample Preprocessing Attrition

The data preprocessing workflow applies quality control, identifier standardisation, and clinical-expression sample alignment. The table below outlines sample retention and attrition rationale for each cohort.

Table 2: Sample Attrition Across Preprocessing Steps

Cohort	Preprocessing Step	N Initial	N Retained	N Removed	Rationale
Liu 2019	Clinical data loaded	122	122	0	Merged patient-level and sample-level clinical records.
Liu 2019	Clinical data harmonised	122	122	0	Applied identifier standardisation, clinical cleaning, baseline filtering, and variable harmonisation.
Liu 2019	Clinical-expression alignment	122	122	0	Retained samples with matching clinical and expression data.
Hugo 2016	Clinical data loaded	27	27	0	Merged patient-level and sample-level clinical records.
Hugo 2016	Clinical data harmonised	27	27	0	Applied identifier standardisation, clinical cleaning, baseline filtering, and variable harmonisation.
Hugo 2016	Clinical-expression alignment	27	27	0	Retained samples with matching clinical and expression data.
Riaz 2017	Clinical data loaded	107	107	0	Merged patient-level and sample-level clinical records.
Riaz 2017	Clinical data harmonised	107	107	0	Applied identifier standardisation, clinical cleaning, baseline filtering, and variable harmonisation.
Riaz 2017	Clinical-expression alignment	107	107	0	Retained samples with matching clinical and expression data.
TCGA-SKCM	Clinical data loaded	448	448	0	Merged patient-level and sample-level clinical records.
TCGA-SKCM	Clinical data harmonised	448	448	0	Applied identifier standardisation and clinical data cleaning.
TCGA-SKCM	Clinical-expression alignment	448	443	5	Retained samples with matching clinical and expression data.

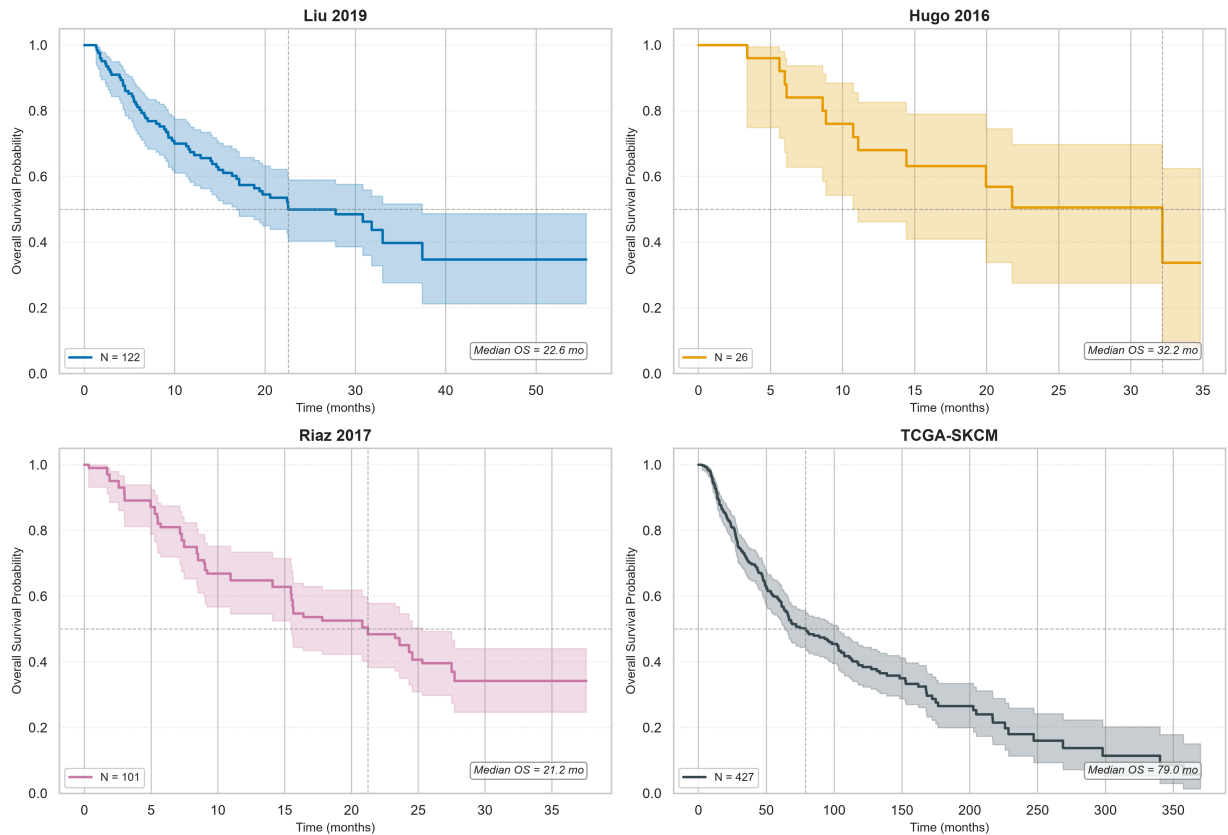
Key Observations

- Cohort Size:** The largest individual cohort is **TCGA-SKCM** with **N = 443** patients. Across all four cohorts, **N = 699** cleaned clinical samples were harmonised.
- Sample Attrition:** Preprocessing quality control and clinical-expression alignment evaluated **704** initial records and removed **5** sample(s) across cohorts (**Liu 2019** retained 100% of samples (N = 122); **Hugo 2016** retained 100% of samples (N = 27); **Riaz 2017** retained 100% of samples (N = 107); **TCGA-SKCM** lost **5** sample(s) (448 → 443)).
- Overall Survival:** Median overall survival was estimated using the Kaplan-Meier survival function. Where the estimated survival probability did not fall below 50% during follow-up, median OS was reported as not reached.
- Follow-up Duration:** **TCGA-SKCM** demonstrated the longest median follow-up duration at **41.6 months**.
- Event Rates:** Observed OS event rates varied between cohorts, reflecting differences in patient composition, disease staging, treatment regimens, follow-up duration, and censoring.
- Variable Completeness:** Demographic (age, sex) and treatment annotations vary in availability across datasets, requiring careful consideration during multi-cohort synthesis.

Overall Survival Curves (KM Plots)

The unstratified Kaplan-Meier overall survival curves for each cohort are shown below. Median overall survival is annotated for each cohort where reached.

Overall Survival — All Cohorts



Overall Survival KM Curves (All Cohorts)

Analysis Notes

Overall survival was analysed using the available cohort-specific survival duration and event-status variables.

Records were excluded from the survival analysis if they had:

- Missing survival time.
- Missing survival event status.
- Non-numeric survival values.
- A survival time less than or equal to zero.

Median overall survival was estimated using the Kaplan-Meier survival function. Where the estimated survival probability did not fall below 50% during follow-up, median OS was reported as **NR (not reached)**.

Treatment percentages use the number of patients with an available treatment annotation as the denominator where this differs from the total cohort size. TCGA treatment-history categories are based on binary treatment-type indicators and are not necessarily mutually exclusive.

Sample attrition tracking records sample filtering from initial cBioPortal data ingestion through clinical-expression alignment.

The analysis is descriptive and unstratified. It does not adjust for demographic, clinical, molecular, treatment, or study-specific confounding factors.